

* Ejercicio 10 = Hallar, si es posible, k / f sea continua en $\mathbb{R}^2$

$$f(x,y) = \begin{cases} \frac{x^3 - x(y+1)^2}{x^2 + (y+1)^2} & \text{si } (x,y) \neq (0,-1) \\ k & \text{si } (x,y) = (0,-1) \end{cases}$$

$\rightarrow f(0,-1) = k \rightarrow$ existe imagen $\rightarrow$ Hallar $k \in \mathbb{R} \mid \lim_{(x,y) \rightarrow (0,-1)} f(x,y) = f(0,-1) = k$

$$\rightarrow \lim_{(x,y) \rightarrow (0,-1)} \frac{x^3 - x(y+1)^2}{x^2 + (y+1)^2} = \text{Ind } \frac{0}{0}$$

• Intentar transformar expresión

$$\frac{x^3 - x(y+1)^2}{x^2 + (y+1)^2} = \frac{x[x^2 - (y+1)^2]}{x^2 + (y+1)^2} = x \left(\frac{x^2 - (y+1)^2}{x^2 + (y+1)^2} \right) \quad \text{División polinómica}$$

$$\downarrow$$

$$x \left[\frac{(x - (y+1))(x + (y+1))}{x^2 + (y+1)^2} \right] \Rightarrow \text{No sirve simplificar expr}$$

$$x \left(\frac{-2(y+1)^2}{x^2 + (y+1)^2} + 1 \right)$$

$$\frac{x^2 - (y+1)^2}{-x^2 + (y+1)^2} \cdot \frac{x^2 + (y+1)^2}{1}$$

$$\frac{0 - 2(y+1)^2}{0 - 2(y+1)^2}$$

$$\lim_{(x,y) \rightarrow (0,-1)} \frac{x^3 - x(y+1)^2}{x^2 + (y+1)^2} = \lim_{(x,y) \rightarrow (0,-1)} x \left(\frac{x^2 - (y+1)^2}{x^2 + (y+1)^2} \right) = \lim_{(x,y) \rightarrow (0,-1)} x \left(\frac{x^2}{x^2 + (y+1)^2} - \frac{(y+1)^2}{x^2 + (y+1)^2} \right)$$

$$= \lim_{(x,y) \rightarrow (0,-1)} \underbrace{\frac{x \cdot x^2}{x^2 + (y+1)^2}}_{\substack{\text{acotado} \\ B^*(0,-1, \epsilon)}} - \lim_{(x,y) \rightarrow (0,-1)} \underbrace{\frac{x(y+1)^2}{x^2 + (y+1)^2}}_{\text{acotado}} = 0 - 0 = 0 \rightarrow \boxed{k=0}$$

DERIVADA FUNCIONES VARIAS VARIABLES

* Ejercicio 10 = Hallar derivadas parciales y evaluar en punto

a) $f(x,y) = xy + x^2$, en $(2,0)$

rectángulo $x \rightarrow \overset{h}{f(2,0) + 4(x-2)}$

• $\frac{\partial f}{\partial x}(x,y) = y + 2x$, $\frac{\partial f}{\partial x}(2,0) = 4$ • $\frac{\partial f}{\partial y}(x,y) = x$, $\frac{\partial f}{\partial y}(2,0) = 2$

b) $f(x,y) = \sinh(x^2 + y)$, en $(1,-1)$

• $\frac{\partial f}{\partial x}(x,y) = \cosh(x^2 + y) \cdot 2x$, $\frac{\partial f}{\partial x}(1,-1) = \cosh(0) \cdot 2 = 2$

• $\frac{\partial f}{\partial y}(x,y) = \cosh(x^2 + y)$, $\frac{\partial f}{\partial y}(1,-1) = \cosh(0) = 1$

c) $f(x, y, z) = \frac{xz}{y+z}$, en $(1, 1, 1)$

• $\frac{\partial f}{\partial x}(x, y, z) = \frac{z}{y+z}$, $\frac{\partial f}{\partial x}(1, 1, 1) = \frac{1}{2}$

• $\frac{\partial f}{\partial y}(x, y, z) = xz \cdot \frac{-1}{(y+z)^2}$, $\frac{\partial f}{\partial y}(1, 1, 1) = -\frac{1}{4}$

• $\frac{\partial f}{\partial z}(x, y, z) = \frac{\partial}{\partial z} \left(\frac{xz}{y+z} \right) = \frac{(x)(y+z) - (1)(xz)}{(y+z)^2} = \frac{xy + xz - xz}{(y+z)^2} = \frac{xy}{(y+z)^2}$, $\frac{\partial f}{\partial z}(1, 1, 1) = \frac{1}{4}$

d) $f(x, y, z) = \ln(1+x+y^2z)$, en $(1, 2, 0)$ y $(0, 0, 0)$

$\frac{\partial f}{\partial x}(x, y, z) = \frac{\partial}{\partial x} (\ln(1+x+y^2z)) = \frac{1}{1+x+y^2z}$, $\frac{\partial f}{\partial x}(1, 2, 0) = \frac{1}{2}$, $\frac{\partial f}{\partial x}(0, 0, 0) = 1$

$\frac{\partial f}{\partial y}(x, y, z) = \frac{1}{1+x+y^2z} \cdot \frac{\partial}{\partial y} (1+x+y^2z) = \frac{2zy}{1+x+y^2z}$, $\frac{\partial f}{\partial y}(1, 2, 0) = 0$, $\frac{\partial f}{\partial y}(0, 0, 0) = 0$

$\frac{\partial f}{\partial z}(x, y, z) = \frac{y^2}{1+x+y^2z}$, $\frac{\partial f}{\partial z}(1, 2, 0) = 2$, $\frac{\partial f}{\partial z}(0, 0, 0) = 0$

e) $f(x, z) = \sin(x\sqrt{z})$, $(\frac{\pi}{3}, 4)$

$\frac{\partial f}{\partial x}(x, z) = \cos(x\sqrt{z}) \sqrt{z}$, $\frac{\partial f}{\partial x}(\frac{\pi}{3}, 4) = (\frac{1}{2})\sqrt{4} = -1$

$\frac{\partial f}{\partial z}(x, z) = \cos(x\sqrt{z}) \frac{x}{2\sqrt{z}}$, $\frac{\partial f}{\partial z}(\frac{\pi}{3}, 4) = \frac{\pi}{3} \cdot (\frac{1}{2}) \cdot \frac{1}{4} = -\frac{\pi}{24}$

f) $f(x, y) = \frac{1}{\sqrt{x^2+y^2}}$, en $(-3, 4)$

• $\frac{\partial f}{\partial x}(x, y) = \left(\frac{1}{\sqrt{x^2+y^2}} \right)' = \frac{-1}{2} \frac{1}{(x^2+y^2)^{3/2}} \cdot 2x$, $\frac{\partial f}{\partial x}(-3, 4) = \frac{-1}{2} \cdot \frac{1}{(9+16)^{3/2}} \cdot -6 = \frac{3}{125}$

• $\frac{\partial f}{\partial y}(x, y) = \frac{-1}{2} \cdot \frac{1}{(x^2+y^2)^{3/2}} \cdot 2y$, $\frac{\partial f}{\partial y}(-3, 4) = \frac{-4}{(25)^{3/2}} = \frac{-4}{125}$

g) $f(x, y) = \int_x^y \sin(\ln(1+t^3)) dt$, en $(1, 2)$

$\int \sin(\ln(1+t^3)) dt \Rightarrow$ Atós:

$$⑧ \quad f(x,y) = \int_x^{y^2} \sin(\ln(1+t^3)) dt, \text{ en } (1,2)$$

$$\exists G(t) / G'(t) = \sin(\ln(1+t^3)) \quad f(x,y) = G(y^2) - G(x)$$

$$\frac{\partial f}{\partial x}(x,y) = \frac{\partial}{\partial x} \left(\underbrace{G(y^2)}_{\in \mathbb{R}} - G(x) \right) = -G'(x) \underbrace{\frac{\partial}{\partial x}(x)}_1 = -\sin(\ln(1+x^3)) \rightarrow \frac{\partial f}{\partial x}(1,2) = -\sin(\ln(1+1)) = -\sin(\ln(2))$$

$$\frac{\partial f}{\partial y}(x,y) = \frac{\partial}{\partial y} \left(G(y^2) - \underbrace{G(x)}_{\in \mathbb{R}} \right) = G'(y^2) \frac{\partial}{\partial y}(y^2) = \sin(\ln(1+y^6)) 2y \rightarrow \frac{\partial f}{\partial y}(1,2) = \sin(\ln(65)) 4$$

* Ejercicio 12 = Analizar derivada parcial en (0,0) y analizar continuidad

$$⑨ \quad f(x,y) = \begin{cases} \frac{2x^3 - y^3}{x^2 + 3y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

• Analizar derivadas parciales en (0,0)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\partial f}{\partial x}(x,y) = 2$$

$$\lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \left[\frac{2h^3 - 0}{h^2 + 0} - 0 \right] \cdot \frac{1}{h} = \lim_{h \rightarrow 0} \frac{2h^3}{h^3} = 2 = \frac{\partial f}{\partial x}(0,0)$$

derivada continua

$$\lim_{h \rightarrow 0} \frac{f(0,h) - f(0,0)}{h} = \lim_{h \rightarrow 0} \left[\frac{0 - h^3}{0 + 3h^2} - 0 \right] \cdot \frac{1}{h} = \lim_{h \rightarrow 0} \frac{-h^3}{3h^3} = -\frac{1}{3} = \frac{\partial f}{\partial y}(0,0)$$

• Analizar continuidad f en (0,0)

$$f(0,0) = 0 \rightarrow \text{existe imagen}$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{2x^3 - y^3}{x^2 + 3y^2} \stackrel{0}{=} \lim_{(x,y) \rightarrow (0,0)} \frac{2x^3 - y^3}{x^2 + 3y^2} = \lim_{(x,y) \rightarrow (0,0)} \frac{x^3 + x^3 - y^3}{x^2 + 3y^2} \quad ?$$

$$\begin{array}{r} 2x^3 - y^3 \quad | \quad x^2 + 3y^2 \\ - 2x^3 + 6y^2x \quad | \quad 2x - \frac{2}{3}y + 2yx \\ \hline 0 - y^3 - 6y^2x \quad | \quad -y^3 - \frac{1}{3}yx^2 \\ - -y^3 - \frac{1}{3}yx^2 \\ \hline 0 - 6y^2x + \frac{1}{3}yx^2 \\ - + 6y^2x + 2yx^3 \\ \hline 0 \quad 2yx^3 + \frac{1}{3}yx^2 \end{array}$$

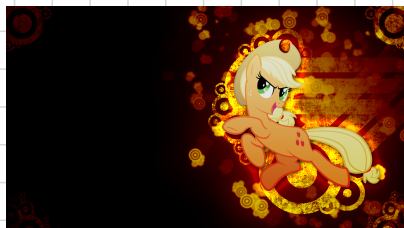
• probar recta $y = mx$

$$\lim_{x,y \rightarrow (0,0)} \frac{x^3(2-m^3)}{x^2(1+3m^2)} = 0 \rightarrow \mathbb{R}$$

KHE

creo que se puede reemplazar por coord polares

$$\begin{cases} x = r \cos(\alpha) \\ y = r \sin(\alpha) \end{cases} \quad \boxed{?} \quad ? ? ? ? ?$$



這一個 不好玩了 ㄟ

想了好久

$$\textcircled{b} \quad f(x,y) = \begin{cases} \frac{x^2 - 2y^2}{x - y} & x \neq y \\ 0 & \text{en cualquier otro caso} \end{cases}$$

- Hallar derivadas parciales en $(0,0)$

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \left[\frac{h^2 - 0}{h - 0} - 0 \right] \cdot \frac{1}{h} = 1$$

$$\frac{\partial f}{\partial y}(0,0) = \lim_{h \rightarrow 0} \frac{f(0,h) - f(0,0)}{h} = \lim_{h \rightarrow 0} \left[\frac{0 - 2h^2}{0 - h} - 0 \right] \cdot \frac{1}{h} = 2$$

- Hallar continuidad en $(0,0)$ de f

- $f(0,0) = 0 \rightarrow$ existe imagen

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - 2y^2}{x - y} \stackrel{\substack{\rightarrow 0 \\ \frac{0}{0}}}{=} \lim_{(x,y) \rightarrow (0,0)} \left[\frac{x^2 - y^2}{x - y} - \frac{y^2}{x - y} \right] = \lim_{(x,y) \rightarrow (0,0)} \frac{(x-y)(x+y)}{\cancel{x-y}} - \lim_{(x,y) \rightarrow (0,0)} \frac{y^2}{x-y}$$